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Introduction

Versatile **software engineer** with experience spanning **web development**, **systems programming**, and **ML / data infrastructure**.

At **i-Scream Media**, built and scaled a production **multi-modal content-understanding platform** – VLM, OCR, and ASR extraction pipelines processing **millions of assets** on on-premise **H200 GPU** infrastructure.

Background in **C/C++ systems programming** (IOCP, multi-threaded architectures, Linux kernel) with prior internship experience in **React/TypeScript** web platforms at NAVER.

Trilingual (Korean, English, Mandarin) with international experience across the US, China, and South Korea.

Education

Shanghai Jiao Tong University

Bachelor of Engineering (September 2018 – March 2025)

University of Michigan – Shanghai Jiao Tong University Joint Institute

- Major: Electrical and Computer Engineering
- Minor: Computer Science
- Minor: Entrepreneurship

Fort Worth Christian School International

March 2015 – December 2017

Pre-University Education

Programming Languages

 C  C++  Python  Java  JavaScript  Rust  SQL

Frameworks

 React  TypeScript  React Native  Next.js

ML & Data Infrastructure

 vLLM  Hugging Face  FastAPI  Docker  CUDA / H200 / RTX 4090  DuckDB
 PostgreSQL  MySQL  Parquet  Streamlit  FFmpeg

Collaboration & Tools

 Git  GitHub  Slack  Jira  LaTeX  Markdown  MkDocs

Systems & Network Programming

- Built a production **VLM image classification pipeline** on **H200 GPU** with **FP8/AWQ quantization** for cost-efficient on-premise inference.
 - Proficient in **Windows System Programming**, focusing on **asynchronous I/O** models such as **IOCP** and **Overlapped I/O**.
 - Experienced with **socket programming (TCP/UDP)** using **Winsock2** and **POSIX sockets**.
 - Designed and benchmarked **multi-threaded, event-driven, and asynchronous** server architectures.
 - Understanding of **packet processing, session management, and concurrency control** for scalable game server systems.
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Experience

i-Scream Media

February 2026 – Present

Junior Researcher, Seoul, South Korea

- Built and scaled an on-premise **multi-modal content-understanding platform** that extracts metadata and curriculum tags from a large children's educational media library, spanning **images, video, documents, interactive HTML, and legacy Flash** assets under a unified metadata contract.
 - Shipped the original **VLM image-safety classifier (Gemma 4 31B, FP8)** on **H200 GPU** via **vLLM** — a **2-pass pipeline** (VLM visual analysis + deterministic profanity/slur text analysis) producing Korean media ratings (all/12/15/18), served via a **FastAPI** production server.
 - Engineered **high-throughput, resumable extraction pipelines** processing **5M+ images and 1.5M+ curriculum records at 99.9%+ success**, using transactional claim queues, checkpointing, and watchdog-based recovery; batching optimizations cut a projected multi-week run to a few days (**~5× throughput**).
 - Built **video and document AI pipelines** — frame/audio extraction with **Whisper** transcription and **VAD**, plus layout-aware OCR (**PaddleOCR-VL**) over PDF/PPT/HWP formats.
 - Designed the platform's **data layer**: stable content-hash identity, content-specific schemas, and an analytical **DuckDB / Parquet** mart benchmarked **50–180× faster** than the operational **MySQL / PostgreSQL** store.
 - Automated **Mac → H200 deployment** (standalone rsync scripts, systemd services, locked `uv` environments) and shipped **Streamlit / static QA dashboards** for coverage and data-quality review.
 - Managed shared multi-GPU infrastructure (**H200 NVL, RTX 4090**) with **FP8/AWQ quantization**, tensor-parallel topologies, and concurrency tuning for cost-efficient inference under children's data-privacy constraints.
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NAVER Start Up Factory – DeepAuto AI

August 2023 – January 2024

Platform Engineer Intern

- Developed a Diffusion Image Generator web platform, enhancing AI-based image synthesis.
- Designed and implemented client-side web features using **React** and **TypeScript**.

- Integrated APIs with backend servers and databases to improve system interoperability.
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Digital Company Inc.

March 2021 – December 2021

Web Development Intern (Remote)

- Developed marketing websites using **TypeScript** and **React**, ensuring modern UI design.
 - Ensured responsive design and cross-browser compatibility, improving user experience.
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Projects

- **AI Image Generator (GenAIOps)** – *Fall 2023* – ML-model visualization & optimization web platform with real-time performance monitoring – [YouTube](#) · *Python, FastAPI, React, Next.js*
- **AI Model Optimization Statistics Viewer** – *Winter 2023* – real-time dashboards for AI-model performance metrics – [YouTube](#) · *TypeScript, React, Redux, ECharts*

Academic projects (SJTU · ECE coursework):

- **Multi-Threading Database** – *Fall 2024* – 64-thread-optimized database with semaphore-based read/write locking – [GitHub](#) · *C++, PThreads, SQLite*
 - **Lottery Scheduling (Linux Kernel)** – *Winter 2024* – eBPF ticket-based scheduler with shared dispatch queues – [GitHub](#) · *C, eBPF, Linux Kernel*
 - **Big Data Million Song** – *Summer 2024* – distributed recommendation over the Million Song Dataset – [GitHub](#) · *Hadoop, Spark, Python*
 - **Web App & Database Design** – *Summer 2024* – capstone-tracking web app with OAuth + decentralized DB – [GitHub](#) · *MySQL, React, Node.js*
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Last updated: July 2026